

THE LAWS & APPLICATIONS OF EXPONENTS

Beginner-Level Master Practice Worksheet

Subject: Mathematics

Grade Level: Class VII (CBSE/ICSE)

Chapter Focus: Basic Powers & Laws of Exponents

Structure: 10 MCQs & 10 Subjective (with HOTS)

Worksheet Objectives & Instructions

Objectives: This worksheet is structured precisely around the foundational syllabus of Class VII. It provides solid reinforcement of exponential notations, evaluations, fractional bases, laws of exponents (product, quotient, power-of-a-power, and zero exponent laws), units conversion, and basic real-world daily life models (paper folding, tournament structures, and bacterial growth).

Structure Instructions:

- **Section A: Multiple Choice Questions (MCQs 1–10):** Direct textbook-relevant applications, including two Higher-Order Thinking Skills (HOTS) questions to stretch conceptual boundaries.
- **Section B: Problem-Based Subjective Questions (Q11–20):** Step-by-step mathematical problems requiring structured solutions, including two beginner-appropriate HOTS questions.
- **Section C: Comprehensive Solution Key:** A complete, detailed answer sheet is included at the end of the document to allow students to verify and learn step-by-step.

Section A: Multiple Choice Questions (MCQs)

Choose the correct option for each of the following questions. Show basic working wherever required.

Q1. Understanding the Components: In the exponential term $(-3)^4$, what are the base and the exponent respectively?

- (a) Base = 3, Exponent = 4
- (b) Base = -3 , Exponent = 4
- (c) Base = 4, Exponent = -3
- (d) Base = -4 , Exponent = 3

Q2. Product Law for Integers: Apply the product law of exponents ($a^m \times a^n = a^{m+n}$) to simplify the expression $2^3 \times 2^2$. The result expressed as a rational number is:

- (a) 12
- (b) 24
- (c) 32
- (d) 64

Q3. Expressing Rational Numbers as Powers: When expressed in exponential form, the rational number $\frac{-1}{128}$ is equal to:

- (a) $\left(\frac{-1}{2}\right)^7$
- (b) $\left(\frac{1}{2}\right)^7$
- (c) $\left(\frac{-1}{2}\right)^6$
- (d) $\left(\frac{1}{2}\right)^6$

Q4. Zero Exponent Law: According to the exponent law $a^0 = 1$ for any non-zero rational number, what is the value of $(-5)^0$?

- (a) -5
- (b) 0
- (c) 1
- (d) -1

Q5. Units of Length Conversion: Convert 1.56 km directly to centimetres (cm) and write the result in standard scientific/exponential notation. (Recall: $1 \text{ km} = 10^5 \text{ cm}$).

- (a) $1.56 \times 10^5 \text{ cm}$
- (b) $1.56 \times 10^7 \text{ cm}$
- (c) $1.56 \times 10^3 \text{ cm}$
- (d) $1.56 \times 10^6 \text{ cm}$

Q6. Quotient Law for Rational Bases: Using the quotient law of exponents $\left(\frac{a}{b}\right)^m \div \left(\frac{a}{b}\right)^n = \left(\frac{a}{b}\right)^{m-n}$, simplify $\left(\frac{3}{5}\right)^7 \div \left(\frac{3}{5}\right)^5$.

- (a) $\frac{9}{25}$
- (b) $\frac{3}{5}$
- (c) $\frac{27}{125}$
- (d) $\frac{6}{10}$

Q7. Standard Scientific Form: The number 346,130 when written in standard exponential/scientific form is equal to:

- (a) 3.4613×10^5
- (b) 3.4613×10^6
- (c) 34.613×10^4
- (d) 3.4613×10^4

Q8. Daily Life Exponential Growth: Under a biology experiment, a bacterial culture begins with exactly 10 single-celled bacteria. If the population doubles every hour (reproducing exponentially as $N \times 2^h$), how many bacteria will be present after 4 hours?

- (a) 40
- (b) 80
- (c) 160
- (d) 320

HOTS Question 1

Q1. Solving Exponential Equations: If $3^{2x-1} + 3^0 = 10$, what is the value of x ?

- (a) 1
- (b) 1.5
- (c) 2
- (d) 3

HOTS Question 2

Q1. Base Decomposition & Substitution: If $2^a \times 3^b = 72$, what is the value of $a^b + b^a$?

- (a) 12
- (b) 17
- (c) 25
- (d) 72

Section B: Problem-Based Questions

Write out complete, step-by-step algebraic working for each of the following questions.

- Q1. Evaluating Negative Rational Powers:** Evaluate and express as a standard rational number in simplest form:

$$\left(\frac{-3}{4}\right)^3$$

- Q2. Prime Exponential Base Matching:** Find the prime factors of 3125 and express the number in exponential notation with a base of 5.

- Q3. Combining Powers of Same Base:** Using the laws of exponents, simplify the following expression and write the final output as a single power:

$$\left(\frac{13}{43}\right)^7 \times \left(\frac{13}{43}\right)^2$$

- Q4. Power of a Power Reduction:** Simplify the nested exponential expression using the law $(a^m)^n = a^{mn}$, writing the output as a single power:

$$\left[\left(\frac{-7}{15}\right)^3\right]^4$$

- Q5. Inverse Quotient Equations:** Solve for the unknown integer value m in the following equation:

$$2^m \div 2^3 = 32$$

- Q6. Length Conversions in Daily Life:** Convert 98 hm (hectometres) directly into centimetres (cm) and write the final value in standard exponential scientific form (e.g., $k \times 10^p$). (Note: 1 hm = 100 m and 1 m = 100 cm).

- Q7. Sports Tournament Modeling:** In a knockout tennis competition, exactly 128 players enter. In each successive round, the losers are eliminated and the winners advance.

- (i) How many players remain after 5 rounds?
- (ii) Express this remaining player count as a power of base 2.

- Q8. Evaluating Mixed Bases:** Using laws of exponents and prime factorization, simplify and evaluate the following expression as a positive integer:

$$\frac{3^4 \times 2^3}{6^2}$$

HOTS Question 3

Q1. Reciprocals & Multi-step Fractional Exponents: If a rational ratio $\frac{p}{q}$ is defined as:

$$\frac{p}{q} = \left(\frac{3}{2}\right)^3 \div \left(\frac{9}{4}\right)^2$$

Find the exact value of the negative exponent term $\left(\frac{p}{q}\right)^{-2}$ as a simplified fraction.

HOTS Question 4

Q1. Paper Folding Progression Analysis: A sheet of poster paper has a thickness of exactly 0.1 mm. Folding the paper in half doubles its thickness.

- (i) Write an exponential mathematical expression representing the thickness of the sheet (in mm) after k folds.
- (ii) Determine the minimum number of folds required so that the overall folded paper thickness is at least 2.5 cm (which is equal to 25 mm). Show your mathematical testing.

Section C: Complete Solution Key & Step-by-Step Explanations

Section A: Multiple Choice Solutions

Q1 Solution

In an exponential term of the form a^n , a is defined as the **base** and n is the **exponent** or index.

- For the term $(-3)^4$, the base is the entire quantity inside the parenthesis, which is -3 .
- The exponent (the power to which the base is raised) is 4.

Correct Answer: (b) Base = -3 , Exponent = 4

Q2 Solution

The Product Law states that for any non-zero base a :

$$a^m \times a^n = a^{m+n}$$

Applying this to the given expression:

$$2^3 \times 2^2 = 2^{3+2} = 2^5$$

Evaluating 2^5 :

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$$

Correct Answer: (c) 32

Q3 Solution

To express the rational fraction $\frac{-1}{128}$ in exponential form:

- Identify the prime factorization of the denominator:

$$128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7$$

- Since $-1 = (-1)^7$ (because a negative base raised to an odd power remains negative):

$$\frac{-1}{128} = \frac{(-1)^7}{2^7} = \left(\frac{-1}{2}\right)^7$$

Correct Answer: (a) $\left(\frac{-1}{2}\right)^7$

Q4 Solution

The Zero Exponent Law states that any non-zero real number or rational fraction raised to the power of zero is exactly equal to 1.

$$a^0 = 1 \quad (\text{where } a \neq 0)$$

Therefore, $(-5)^0 = 1$. **Correct Answer:** (c) 1

Q5 Solution

We convert kilometres to centimetres step-by-step:

$$1.56 \text{ km} = 1.56 \times 1000 \text{ m} = 1560 \text{ m}$$

$$1560 \text{ m} = 1560 \times 100 \text{ cm} = 156,000 \text{ cm}$$

To write 156,000 cm in standard scientific exponential notation, we place the decimal point after the first non-zero digit:

$$156,000 = 1.56 \times 10^5 \text{ cm}$$

Correct Answer: (a) $1.56 \times 10^5 \text{ cm}$

Q6 Solution

According to the Quotient Law for identical bases:

$$\left(\frac{a}{b}\right)^m \div \left(\frac{a}{b}\right)^n = \left(\frac{a}{b}\right)^{m-n}$$

Applying this to our expression:

$$\left(\frac{3}{5}\right)^7 \div \left(\frac{3}{5}\right)^5 = \left(\frac{3}{5}\right)^{7-5} = \left(\frac{3}{5}\right)^2$$

Evaluating the power:

$$\left(\frac{3}{5}\right)^2 = \frac{3^2}{5^2} = \frac{9}{25}$$

Correct Answer: (a) $\frac{9}{25}$

Q7 Solution

In standard scientific form, a positive real number is written as $a \times 10^n$, where $1 \leq a < 10$ and n is an integer.

- For the number 346,130, the decimal point is implicitly at the end: 346130.0
- We shift the decimal point 5 places to the left to get 3.4613.
- Since we shifted the decimal left by 5 places, the exponent of 10 is +5.

$$346,130 = 3.4613 \times 10^5$$

Correct Answer: (a) 3.4613×10^5

Q8 Solution

We model the exponential growth of the bacteria:

- Initial count (at hour 0): $N = 10$
- General expression after h hours: $P(h) = N \times 2^h = 10 \times 2^h$
- Setting $h = 4$:

$$P(4) = 10 \times 2^4 = 10 \times 16 = 160 \text{ bacteria}$$

Correct Answer: (c) 160

HOTS Q9 Solution

Given the exponential equation:

$$3^{2x-1} + 3^0 = 10$$

Substitute $3^0 = 1$ (by the Zero Exponent Law):

$$3^{2x-1} + 1 = 10$$

Isolate the exponential term:

$$3^{2x-1} = 10 - 1 \implies 3^{2x-1} = 9$$

Express 9 as a power of base 3:

$$3^{2x-1} = 3^2$$

Since the bases are equal, we equate the exponents:

$$2x - 1 = 2$$

Solve the linear equation for x :

$$2x = 3 \implies x = \frac{3}{2} = 1.5$$

Correct Answer: (b) 1.5

HOTS Q10 Solution

We are given:

$$2^a \times 3^b = 72$$

First, find the prime factorization of 72:

$$72 = 2 \times 36 = 2 \times 2 \times 18 = 2 \times 2 \times 2 \times 9 = 2^3 \times 3^2$$

Comparing the terms:

$$2^a \times 3^b = 2^3 \times 3^2$$

By matching the base-2 and base-3 prime factors, we find:

$$a = 3 \quad \text{and} \quad b = 2$$

We are asked to find the value of $a^b + b^a$:

$$a^b + b^a = 3^2 + 2^3 = 9 + 8 = 17$$

Correct Answer: (b) 17

Section B: Subjective Problem Solutions**Q11 Solution**

To evaluate the fractional power $\left(\frac{-3}{4}\right)^3$:

- Apply the power of a quotient law, which states $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$:

$$\left(\frac{-3}{4}\right)^3 = \frac{(-3)^3}{4^3}$$

- Compute the numerator (odd power of a negative number remains negative):

$$(-3)^3 = (-3) \times (-3) \times (-3) = -27$$

- Compute the denominator:

$$4^3 = 4 \times 4 \times 4 = 64$$

- Combine to form the final rational number:

$$\left(\frac{-3}{4}\right)^3 = -\frac{27}{64}$$

Q12 Solution

To express 3125 in exponential form with base 5, perform repeated division by 5:

$$3125 \div 5 = 625$$

$$625 \div 5 = 125$$

$$125 \div 5 = 25$$

$$25 \div 5 = 5$$

$$5 \div 5 = 1$$

Since we divided by 5 exactly five times:

$$3125 = 5 \times 5 \times 5 \times 5 \times 5 = 5^5$$

Correct Answer: 5^5

Q13 Solution

Applying the Product Law for rational bases, which states:

$$\left(\frac{a}{b}\right)^m \times \left(\frac{a}{b}\right)^n = \left(\frac{a}{b}\right)^{m+n}$$

Using the common base $\frac{13}{43}$ and exponents $m = 7, n = 2$:

$$\left(\frac{13}{43}\right)^7 \times \left(\frac{13}{43}\right)^2 = \left(\frac{13}{43}\right)^{7+2} = \left(\frac{13}{43}\right)^9$$

Q14 Solution

Applying the Power of a Power Law, which states:

$$(x^m)^n = x^{m \times n}$$

Here, the base is $\left(\frac{-7}{15}\right)$ and the exponents are 3 and 4:

$$\left[\left(\frac{-7}{15}\right)^3\right]^4 = \left(\frac{-7}{15}\right)^{3 \times 4} = \left(\frac{-7}{15}\right)^{12}$$

Note: Since the power 12 is even, this is also mathematically equivalent to $\left(\frac{7}{15}\right)^{12}$. Both forms are correct.

Q15 Solution

Given the equation:

$$2^m \div 2^3 = 32$$

Apply the Quotient Law ($2^a \div 2^b = 2^{a-b}$) to the left-hand side:

$$2^{m-3} = 32$$

Write 32 as a power of base 2:

$$32 = 2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

Equating the terms:

$$2^{m-3} = 2^5$$

Since bases are identical, we equate the exponents:

$$m - 3 = 5 \implies m = 5 + 3 = 8$$

Correct Answer: $m = 8$

Q16 Solution

First, identify the unit conversion factors to centimetre:

$$1 \text{ hm} = 100 \text{ m}$$

$$1 \text{ m} = 100 \text{ cm}$$

$$\Rightarrow 1 \text{ hm} = 100 \times 100 \text{ cm} = 10,000 \text{ cm} = 10^4 \text{ cm}$$

Now, convert 98 hm:

$$98 \text{ hm} = 98 \times 10^4 \text{ cm} = 980,000 \text{ cm}$$

To write this in standard scientific exponential form ($a \times 10^n$ where $1 \leq a < 10$):

$$980,000 = 9.8 \times 10^5 \text{ cm}$$

Q17 Solution

We model the knockout tournament progression:

- In a knockout tournament, half the players are eliminated each round, meaning the player count is divided by 2 per round.

- After r rounds, the remaining player count is:

$$P(r) = \frac{\text{Initial Players}}{2^r}$$

- Substituting Initial Players = 128 and $r = 5$:

$$P(5) = \frac{128}{2^5}$$

- Write 128 as a power of 2: $128 = 2^7$.

- Apply quotient rule:

$$P(5) = \frac{2^7}{2^5} = 2^{7-5} = 2^2$$

- Evaluating:

$$2^2 = 4 \text{ players}$$

Correct Answer: (i) 4 players, (ii) 2^2 players

Q18 Solution

To evaluate the expression $\frac{3^4 \times 2^3}{6^2}$:

- Prime factorize the base of the denominator: $6 = 2 \times 3$.
- Rewrite the denominator using the law $(ab)^n = a^n \times b^n$:

$$6^2 = (2 \times 3)^2 = 2^2 \times 3^2$$

- Substitute back into the expression:

$$\frac{3^4 \times 2^3}{2^2 \times 3^2}$$

- Regroup the same bases and apply the Quotient Law:

$$(3^{4-2}) \times (2^{3-2}) = 3^2 \times 2^1$$

- Evaluate the final positive integer:

$$3^2 \times 2 = 9 \times 2 = 18$$

Correct Answer: 18

HOTS Q19 Solution

We are given the equation:

$$\frac{p}{q} = \left(\frac{3}{2}\right)^3 \div \left(\frac{9}{4}\right)^2$$

First, expand both fractional terms:

$$\begin{aligned}\left(\frac{3}{2}\right)^3 &= \frac{3^3}{2^3} = \frac{27}{8} \\ \left(\frac{9}{4}\right)^2 &= \frac{9^2}{4^2} = \frac{81}{16}\end{aligned}$$

Substitute back and perform the fractional division (multiply by the reciprocal):

$$\frac{p}{q} = \frac{27}{8} \div \frac{81}{16} = \frac{27}{8} \times \frac{16}{81}$$

Simplify the fraction multiplication by cross-cancelling:

$$\frac{p}{q} = \left(\frac{27}{81}\right) \times \left(\frac{16}{8}\right) = \left(\frac{1}{3}\right) \times 2 = \frac{2}{3}$$

We need to find the value of the term with a negative exponent:

$$\left(\frac{p}{q}\right)^{-2} = \left(\frac{2}{3}\right)^{-2}$$

Apply the negative reciprocal power law, which states $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$:

$$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \frac{3^2}{2^2} = \frac{9}{4}$$

HOTS Q20 Solution

We analyze the paper folding doubling progression:

(i) **Thickness Formula:**

- Initial thickness (0 folds): 0.1 mm
- Thickness after 1 fold: $0.1 \times 2 = 0.2$ mm
- Thickness after 2 folds: $0.1 \times 2^2 = 0.4$ mm
- Therefore, the thickness after k folds in terms of a power of 2 is:

$$T(k) = 0.1 \times 2^k \text{ mm}$$

(ii) **Solving for minimum folds:** We want the thickness to be at least 2.5 cm. Let us convert this target unit to millimetres to match our equation:

$$2.5 \text{ cm} = 2.5 \times 10 \text{ mm} = 25 \text{ mm}$$

Set up the inequality:

$$0.1 \times 2^k \geq 25$$

Divide both sides by 0.1 (or multiply by 10):

$$2^k \geq 250$$

Now, evaluate powers of 2 to find the smallest integer k that satisfies this condition:

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128 \quad (\text{Less than } 250)$$

$$2^8 = 256 \quad (\text{First value greater than or equal to } 250)$$

Therefore, the minimum number of folds required is 8 folds (resulting in a thickness of $0.1 \times 256 = 25.6$ mm, which exceeds 25 mm).